A Large-Scale Workflow for Extracting and Visualizing Genetic Variant Impacts from Alzheimer's Disease Literature

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Introduction

AD and related dementias have a large literature base, but variant-impact evidence remains buried in unstructured abstracts.

Researchers need to know which variants are reported, what effects they have, and which sentence supports each claim.

A scalable eMIND workflow moves beyond small curated abstract sets by repeatedly processing much larger article collections.

The final output links papers, evidence sentences, mutations, genes, disease context, impact type, and identifiers.

Research Questions

Can a large-scale eMIND workflow collect, normalize, and visualize genetic variant impacts from Alzheimer's disease literature?

- Extract mutation-impact relations from PubMed-linked abstracts.
- Classify effects on disease, protein function, biological process, and related phenotypes.
- Retain evidence sentences and metadata such as negation, study-sentence status, mutation inference, and normalization status.
- Make the dataset searchable through eMIND Visuals.

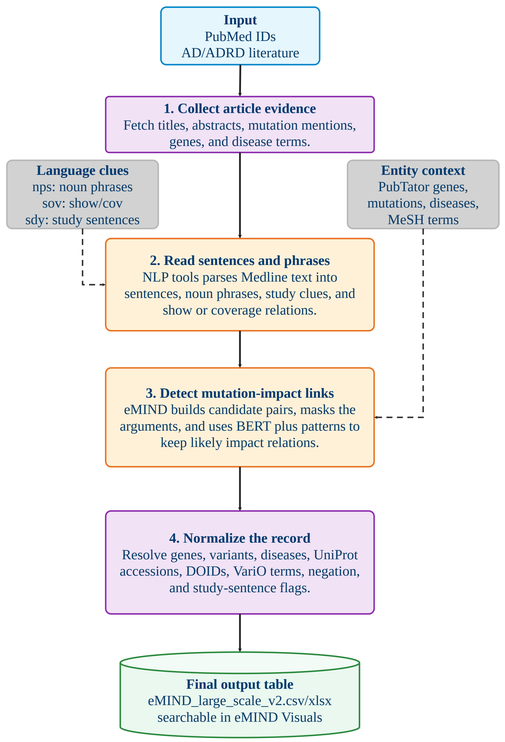
Central Takeaway

The workflow turns a growing, text-heavy AD literature into structured variant-impact records that can be searched, inspected, and traced back to source sentences.

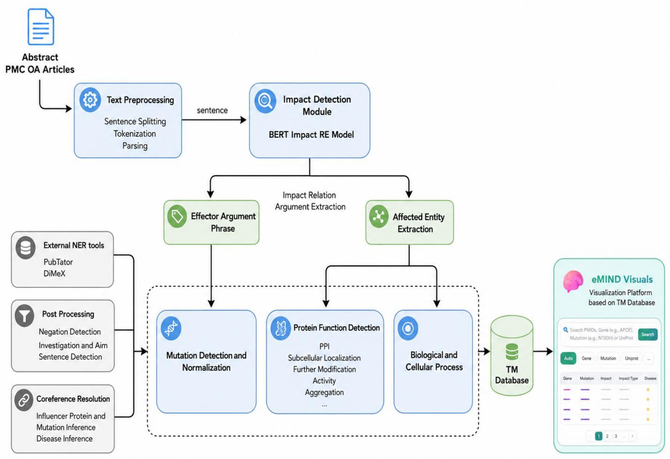
- Automated extraction supplies breadth.
- Post-processing supplies normalization.
- Visualization supplies usability for researchers and curators.

Large-Scale Workflow

The workflow converts AD literature into evidence-linked variant-impact records that a curator can search and verify.



Under the hood: the original eMIND extraction architecture combines preprocessing, BERT relation extraction, external NER, post-processing, and conference resolution.



Structured Output

Each final row keeps the extraction auditable: a source PMID and evidence sentence are carried forward with normalized mutation, gene, disease, and impact fields.

Mutation	mutation_str, normalized ID, RS/HGVS/tmVar
Gene	NCBI Gene ID, symbol, UniProt accession
Impact	Impact_Type, VariO Term, VariO ID
Disease	Disease mention, MESH ID, DOID
Evidence	PMID, sentence index, sentence text
Metadata	negation, study-sentence flag, mut_infer

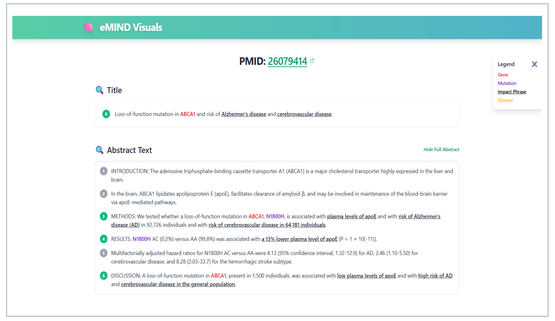
Prior eMIND relation extraction
Recall ~0.84 | Precision ~0.94

Example Result View

The web interface exposes extracted records as a table and keeps every assertion tied to highlighted text in the source abstract.

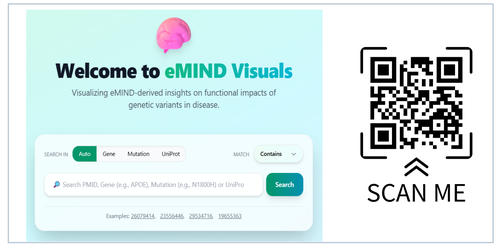

Tool - eMind

Gene	Mutation	Impact	Impact Type	Disease	Sentence
ABCA1 (O95477)	p.Asn1800His	Alzheimer's disease	Disease	Alzheimer's disease (DOID:10652)	0
ABCA1 (O95477)	p.Asn1800His	cerebrovascular disease	Disease	cerebrovascular disease (DOID:6713)	0
ABCA1 (O95477)	p.Asn1800His	risk of Alzheimer's disease (AD)	Disease Risk	Alzheimer's disease (DOID:10652)	3
ABCA1 (O95477)	p.Asn1800His	plasma levels of apoE	Effect on protein abundance	—	3
ABCA1 (O95477)	p.Asn1800His	risk of cerebrovascular disease in 64,181 individuals	Disease Risk	cerebrovascular disease (DOID:6713)	3
ABCA1 (O95477)	p.Asn1800His	a 13% lower plasma level of apoE	Effect on protein abundance	—	4
ABCA1 (O95477)	p.Asn1800His	high risk of AD	Disease Risk	Alzheimer's disease (DOID:10652)	6
ABCA1 (O95477)	p.Asn1800His	cerebrovascular disease in the general population	Disease	cerebrovascular disease (DOID:6713)	6
ABCA1 (O95477)	p.Asn1800His	low plasma levels of apoE	Effect on protein abundance	—	6



Color cues separate genes, mutations, diseases, and impact phrases so users can inspect the extracted relation in context.

eMIND Visuals



- Search by gene, mutation, UniProt accession, or PubMed ID.
- Review a structured table of variant-impact records.
- Open the linked abstract view to inspect highlighted evidence.
- Use filtering and context to support curation and exploratory analysis.

What This Adds

Scale
Reusable pipeline for repeated analysis on larger article collections.

Normalization
Genes, variants, diseases, UniProt accessions, DOIDs, and VariO terms are resolved where possible.

Interpretability
Users see both the structured extraction and the sentence-level evidence behind it.

Conclusion

- Large-scale extraction can make AD variant-impact evidence easier to compare across papers.
- eMIND Visuals turns the processed dataset into a searchable, evidence-linked resource.
- Future work: evaluate at scale, expand article collections, and refine ontology mappings.

References

[1] Gupta, S., Qin, X., Wang, Q., Cowart, J., Huang, H., Wu, C. H., ... & Arighi, C. N. (2023). eMIND: Enabling automatic collection of protein variation impacts in Alzheimer's disease from the literature.bioRxiv, 2023-09.

Acknowledgements

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